

Brand Equity Measurement Models: A Systematic Review

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Abstract

This systematic review examines how different components of brand equity models, particularly Consumer-Based Brand Equity (CBBE), effectively measure and explain brand value across industries. The initial search was conducted electronically in Scopus, Web of Science, ScienceDirect, JSTOR, EBSCOhost, ProQuest, and Google Scholar using keywords such as “brand equity” or “value” and “consumer” and “customer-based brand equity”. Analysis of 24 peer-reviewed articles confirms that four key dimensions, brand awareness, brand associations, brand loyalty, and perceived quality, serve as reliable measures of brand equity across different industry sectors. The study also revealed a structural relationship between the brand equity dimensions. This study identifies the core components of brand equity that organisations can leverage to strengthen their brand positioning and competitive advantage. It explained the relationship between the components of the key brand equity dimensions that would guide firms in developing their brand equity approach to advance their brand competitiveness in the market.

Keywords: brand equity, CBBE, brand identity, brand loyalty, brand awareness, brand associations, perceived quality

1. Introduction

Contemporary marketing strategies focus on building strong consumer relationships and brand differentiation to enhance market competitiveness. This shift has led companies to develop distinctive brand identities as a key differentiating factor in competitive markets. Marketing campaigns are currently aimed at establishing solid brands alongside powerful brand equity to enhance market competitiveness effectively. A strong brand shapes a business's marketing strategies and guarantees high brand equity, enabling firms to relish higher margins, greater customer satisfaction and loyalty, and enhanced cooperation. This, therefore, makes it essential for brands to conceptualise equity models and measures.

Brand equity, a fundamental concept in this respect, has been defined by researchers in various ways. Aaker refers to it as “a set of brand assets and liabilities linked to a brand, its name and symbol that add to or subtract from the value provided by product or service to a firm and its customers” (Aaker, 1991). On the other hand, Keller (1993, p.1) defines brand equity as “the differential effect of brand knowledge on consumer response to the brand's marketing”. Yoo and Donthu similarly define it as “a consumer's different response between a focal brand and an unbranded product when both have the same level of the marketing stimuli and product attributes” (Yoo & Donthu, 2001). Regardless, there seems to be a consensus on brand equity, simplifying it as the incremental utility or value added to a product by the brand's name (Farquhar, 1989).

Previously, Keller and Lehmann stated that brand equity could be measured from corporate or financial-focused brand equity (FBBE) and perception or consumer-based brand equity (CBBE), tentatively the primary brand equity models (Keller & Lehmann, 2006; 2003). However, the FBBE exhibited limitations in terms of providing an unbiased estimation of a brand's intrinsic value merely on business finances or accounting value (Aaker, 1991; Keller, 2010). Hence, the CBBE approach has become much preferable in marketing research due to its capacity to accord businesses a strategic vision of consumer behaviors to develop brand strategies appropriately (Kim et al., 2008; Prasad & Dev, 2000), particularly by positively influencing the brand's equity on customers' brand choice and commitment (Cobb-Walgren et al., 1995), firm's financial performance (Tolba & Hassan, 2009), and brand extension (Pitta & Katsanis, 1995). Aaker (1991) identified the four major dimensions of CBBE: brand awareness, perceived quality, brand loyalty and brand association, which constitute brand equity measurements.

Accordingly, these perceptual dimensions from Aaker (1996), Keller (2010) and Yoo and Donthu (2001) form the

framework of a scale to measure brand equity. However, reservations are made on the capacity of this model to fully and sufficiently explain brand equity since the CBBE model centres around Keller's and Aaker's concept, mainly because Keller, albeit giving a framework to measure the dimensions, fails to provide a nexus between brand awareness and brand equity as core measures of brand equity. Aaker, on similar accounts, only indicates the items contributing to brand equity but does not explain which items are combined to attribute brand equity. However, Nam et al.'s model does not recognise brand awareness, a crucial equity dimension, per Aaker's (1991) and Keller's (1993) models.

The current study concerns whether these dimensions are exhaustive or whether other crucial measurement elements would reliably and validly exhibit brand quality. In particular, the choice of items measuring brand equity is arbitrary due to the need for more literature on the extent to which the existing dimensions contribute to brand equity attainment. This review investigates the extent to which the various dimensions or elements identified by various articles exhibit a constructive and comprehensive assessment, explanation or meaning of brand equity as conceptualised by existing brand equity models, including CBBE.

2. Method

We conducted this systematic review following established PRISMA guidelines to ensure comprehensive and transparent reporting.

2.1 Search Strategy

To search published research in English without regard to the publishing timeframe, we utilise seven comprehensive separate databases. We also searched these databases to identify any existing systematic reviews on brand equity measurement to avoid duplication and identify research gaps. The systematic search was conducted between March and May 2024, covering publications from 1990 to December 2023. Using Endnote software, we merged findings from several databases and eliminated duplicate results using the following search terms: "brand equity" OR "consumer-based brand equity" OR "CBBE" AND "measurement" OR "scale" combined with "brand awareness", "brand loyalty", "perceived quality," and "brand associations". Final search validation was completed on May 10, 2024. Search strategies are listed in Supplementary file 1.

2.2 Studies Selection

Duplicates were removed, and every article was extracted and saved in its own EndNote library (ENDNOTE, 2015). Two separate reviewers chose which studies to include. While reviewer 2 approved papers based on the data and resolved any arguments on any included research, reviewer 1 reviewed abstracts and titles twice, independently. After being thoroughly reviewed by reviewers, the publications were selected for inclusion based on the inclusion and exclusion criteria listed in Table 1 to determine if they contained the pertinent data for the systematic review.

Table 1. Inclusion and exclusion criteria

Question Element	Inclusion criteria	Exclusion criteria
Type of Study	Quantitative studies, cohort studies, case-control studies, cross-sectional studies, and qualitative studies.	Opinion pieces, editorials, letters to the editor, book reviews, or purely theoretical papers without empirical data.
Activity	Studies that develop, test, or analyse models or frameworks for measuring brand equity and its components.	Studies that focus solely on branding strategies without addressing measurement models or their components.
Population	Studies targeting businesses, brands, or industries across any sector, including consumer-based and corporate brand equity models.	Studies irrelevant to brand equity or those exclusively targeting non-business contexts (e.g., personal branding, political branding).
Study Language	Studies published in English.	Studies published in languages other than English.

A Microsoft® Excel Spreadsheet was used to extract and store data and records (Microsoft, Inc., Redmond, Wash., USA).

2.3 Risk Bias Assessment

The Newcastle Ottawa Scale (NOS) was used to assess the risk bias of the included studies. Studies were given low, medium, and high scores according to the following criteria: bias in the selection process, bias in interventions, bias in departures from interventions, bias resulting from missing data, bias in outcomes, and bias in results. Using inclusion and exclusion criteria, preference for selection was rated. Performance bias was assessed by outlining a control arm and taking allocation concealment into account. Biased reporting, selective reporting, data management, and full industry sponsorship were ranked differently. At many meetings, reviewers looked at eligibility restrictions and reporting uniformity. Before choosing a research topic, a second reviewer took into account any discrepancies in the reviewers' ratings.

3. Results

3.1 Search Results

The search process results are summarised in the PRISMA flow chart below (Fig.1). The search strategy yielded a cumulative of 6749 publications. Four more sources were identified via reference or bibliography list tracking and included in the review. After de-duplication, 3412 articles were left, amongst which 2197 records were excluded after the title and abstract screening. Of the 103 articles subjected to full-text screening, 79 were further excluded. Only 24 articles that report brand equity dimensions and allied measurements were considered and included in this review. Twenty-four articles were included in the final analysis, as the study specifically focused on empirical studies using surveys or questionnaire-based data collection methods to measure brand equity components.

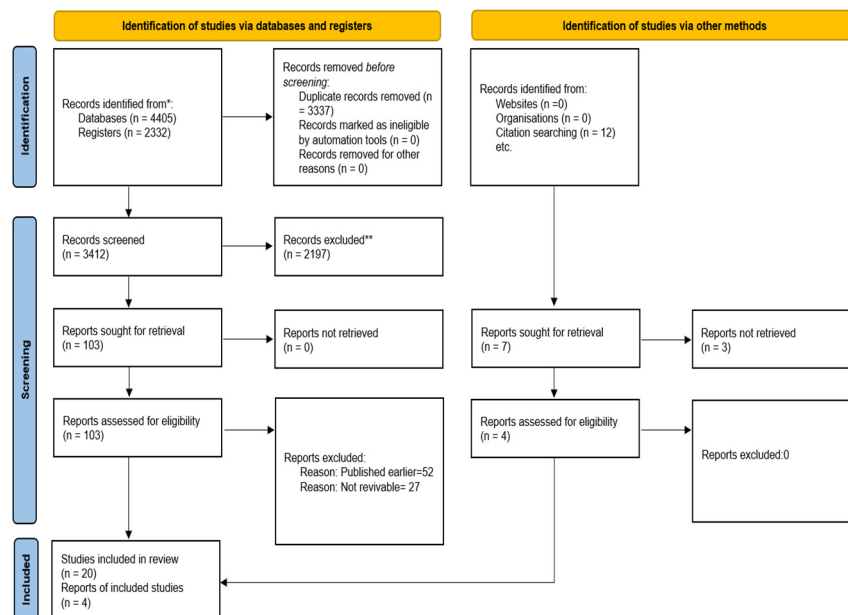


Figure 1. PRISMA flow diagram

3.2 Risk Bias Assessment

Table 2 illustrates the Newcastle-Ottawa Scale (NOS) risk of bias assessment. Nine of the twenty-four studies had a low-risk bias, four had a high-risk bias, and the remaining studies had intermediate-risk bias. Some studies have methodological flaws that stem from the manner they selected their controls. Additionally, a moderate risk bias assessment in the overall research may result from the inclusion of the review (Table 2).

GRADEpro GDT indicated that the included studies in this systematic review showed moderate-quality risk bias. The inclusion of a cohort study, which increases the risk of bias since it cannot randomise the exposure, and the unequal nature of the study were the main causes of the moderate risk bias assessment.

Table 2. Risk of bias assessment in the studies included in this systematic review

Study	Selection				Comparability		Exposure		
	1.	2.	3.	4.	1.		1.	2.	3.
Muhammad & Muath, 2018.	★	★			★		★	★	
Vera & Trujillo, 2017.	★	★					★	★	★
Juga et al., 2018.	★	★							★
Chen & Tseng, 2010.	★	★			★★		★	★	★
Han et al., 2015.	★	★			★		★	★	★
Shabbir et al., 2017)	★	★			★★			★	★
Shariq, 2019.	★	★		★	★★		★	★	★
Bose et al., 2020.	★	★	★		★			★	★
Jeon & Yoo, 2021.	★	★			★★		★	★	★
Kressmann et al., 2006.	★	★						★	
Chen et al., 2019)	★	★			★★		★	★	★
Choi et al., 2017	★	★			★		★	★	
Mabkhot et al., 2017.	★	★					★	★	★
Ong et al., 2015.	★	★							★
Qiao et al., 2022.	★	★			★★		★	★	★
Poonam & Sharma, 2023.	★	★			★		★	★	★
Dada, 2021)	★	★			★★			★	★
Naggar & Bendary, 2017.	★	★		★	★★		★	★	★
Menon, 2019.	★	★	★		★			★	★
Sürücü et al., 2019.	★	★			★★		★	★	★
Hurriyati & Setiawan, 2017.	★	★						★	
Şahin et al., 2011.	★	★		★	★		★	★	★
Netemeyer et al., 2004.	★	★			★★		★	★	★
Ebrahim, 2020.	★	★		★	★★		★	★	★

Notes. Rating scale: 7 to 9 stars = low risk of bias; 4 to 6 stars = moderate risk of bias; 0 to 3 stars = high risk of bias

3.3 Characteristics of Included Studies

Table 3 shows a summary and characteristics of all the included studies. The included studies were mainly from the developing states, except for a few from developed nations. Qualitative and quantitative studies, with a survey as the main methodology design. Few studies included marketing officers and managers in their sample population. However, most data were collected from the primary consumers of that specific brand. The brand contexts evaluated were multiple: food products and food service brands, brewery brands, transport consumers (airline services), telecommunication services, fast-moving consumer goods, automobile customers, SMEs, fashion products, social media brands, and mixed-branded businesses.

Table 3. Characteristics and summary of included studies

Author, Year	Country	Study design	Population	Product category	Brand equity definition/ Attributed model	Measurements/ constructs/ dimensional scales		Brand element/ antecedents	Key findings/ Relationship of consumer value	
Muhammad & Muath, 2018.	Jordan	Quantitative research; Survey	Jordanian marketing officer of sampled corporations	Conceptual	Aaker's model (Aaker, 1996; 1991)	Brand awareness, association, loyalty, perceived quality, and identity	Consumer-based Brand names, URLs, Logos and symbols, Characters, Slogans,	Financial based Packaging	Others N/A	Except for jiggle and character, all the assessed brand elements had statistical significance in the subfields, including brand awareness, loyalty, association, perceived

							Jingles and Signage				quality, and other brand assets, which impact brand equity.
											Perceived brand value had the highest impact on loyalty measurement.
Vera & Trujillo, 2017.	Mexico	Survey	Mexican consumers	Laptops, Women's dress shoes, running shoes, lipstick, canned soda and bottled water	N/A	Brand loyalty (cognitive, affective, intentional and action).	Consumer involvement and customer satisfaction.	Perceived brand value (consumer brand equity)	N/A		However, self-identification with the brand (self-congruence) and perceived equity were significant variables affecting loyalty measurement.
Juga et al., 2018.	Finland	Survey	B2B customers	Brewery company	Service loyalty model in logistics outsourcing relationships by Juga et al. (2018)	Perceived service quality	Value-adding services (operational logistics of service quality)	N/A	N/A		Value-adding services, such as the operational dimension of logistics service quality, bolstered the company's brand equity.
Chen & Tseng, 2010.	Taiwan	Survey	Airline Company	Taiwanese air passenger	Brand equity defined	Brand awareness, Perceived quality, attitudes, Brand loyalty	Marketman's experience, communication strategy, good service quality, recommend	Purchase willingness to repurchase or recommend	Positive, effective perceptions and attitudes		Brand loyalty, mediated by perceptual and behavioural constructs, is the primary measure of brand equity.
Han et al., 2015.	China	Survey	Luxury hotels and chain restaurants	N/A	Consumer-based chain restaurant brand equity (CBCRBE) model	Food and service quality, brand affect, brand awareness, and brand association	Brand reputation and trust (food freshness, hygiene)	N/A	Brand reputation (symbolic consumption)		Brand reputation partially mediates the effects of brand service quality, brand effect, and awareness of brand trust. It also mediates the effect of brand association on brand trust.
Shabbir et al., 2017)	Pakistan	Survey	Telecom sector (Mobilink, Ufone, Telenor, Warid, Zong, SCom)	Pakistani nationals using known telecommunication brands	Aaker's CBBE model(Aaker, 1991)	Brand awareness	Brand image (relations based on customer psyche)	Brand loyalty (repurchase of product)	N/A		Brand awareness effectively mediates the effect of brand image and brand loyalty on brand equity, making it an effective measure in the telecom industry.
Shariq, 2019.	United Arab Emirates	Quantitative survey	Fast-moving consumer goods (FMCG) category	Consumers living in Dubai and Sharjah, and a members of the non-Arab expatriate community	Aaker's Brand Equity Model(Aaker, 1991), Keller's Customer-Based Brand Equity Model (CBBE) (Keller, 1993), and Kapferer's Brand Platform (Kapferer, 2012).	Brand awareness, loyalty, perceived quality and brand associations	Advertising and distribution intensity	Product prices and Promotions	Non-price promotions		Unlike pricing and advertising, distribution intensity and non-price promotions mediated all four brand equity dimensions, particularly brand awareness, ultimately impacting brand equity, hence a

		Mixed study: focus group discussion, in-depth interviews and survey													critical measure.
Bose et al., 2020.	West Bengal (India) and Bangladesh														Psychometric properties, such as brand engagement and brand salience, are instrumental facets of CBBE's measure in the context of place, influencing brand awareness, perceived quality, and loyalty.
															The brand experience influences perceived value (equity) and loyalty through perceived quality, brand association/image and brand awareness.
Jeon & Yoo, 2021.	South Korea	Survey	Grocerants (Food-service industry)	Foodservice consumers with experience of using seven South Korean grocerants	CBBE	Fournier (Madden et al., 2006) and Aaker models (Aaker, 2004)	Brand loyalty	Perceived quality, brand awareness, brand association/image	Brand experience	N/A	N/A				Self-congruity indirectly influences brand loyalty through product involvement and functional congruity, thus impacting customer brand equity.
Kressmann et al., 2006.	Germany	Survey	Automobiles (cars)	Automobile consumers											Airline service quality positively influences repurchase intentions by improving brand awareness and, ultimately, brand value or equity.
Chen et al., 2019)	China	Survey	Airline industry	Airline service consumers.	Tasci(Tasci, 2020) and Keller's(Keller, 2010) model concept										However, brand attractiveness and experience had a non-significant impact on brand awareness and perceived value of brand equity.
Choi et al., 2017	N/A	Survey	Coffeehouse brands	Coffeehouse customers	N/A										Brand experience and personality traits directly impact brand prestige, influencing brand-perceived quality and attitudinal loyalty, subsequently influencing brand consumer equity.
Mabkhot et al., 2017.	Malaysia	Survey	Automobile brands	customers of local automobile brands in Malaysia	Aaker's(1997)model conceptualization										Brand trust mediates the relationship between brand personality and brand loyalty, but does not affect the association between brand image and brand loyalty.
Ong et al., 2015.	Malaysia	Survey	Malay small and medium enterprises (SME) food	Diners of local SME restaurants	Aaker's(1997) conceptualization	CBBE									Brand experience influences attitudinal and behavioural loyalty, which are measurement

				industry brands								constructs of customer equity; however, brand personality only mediates behavioural loyalty. Product perceived value positively influences brand resonance and customer affective commitment; hence, affective commitment and brand resonance effectively mediate the influence of perceived value on brand equity.
Qiao et al., 2022.	China	Survey	Branded products	Branded products consumers	N/A		Product perceived value, brand resonance	Consumer affective commitment	N/A	N/A		Perceived quality significantly drives brand equity within the product category under investigation.
Poonam & Sharma, 2023.	India	Survey	Personal care products	Indian care product consumers	Yoo & Donthu's (2001) model conceptualisation		Brand awareness, loyalty, perceived quality and overall brand quality	Brand equity				Brand associations, brand loyalty and brand image positively affect brand equity.
Dada, 2021)	N/A	Cross-sectional descriptive survey	Conceptual	N/A	N/A		Brand loyalty, brand association	Brand image	N/A	N/A		Brand awareness was disregarded to enhance the association between brand trust and loyalty, and the study noted no correlation between brand experience and brand awareness, association, or perceived quality. However, brand experience through association and perceived quality enhances brand loyalty through trust to impact customer brand equity.
Naggar & Bendary, 2017.	Egypt	Survey	Telecom services sector	Mobile operator subscribers	CBBE conceptualisation model		Brand association, perceived quality, brand awareness and brand loyalty.	Brand experience, brand trust.	N/A	N/A		Brand awareness and brand image substantiate building brand salience and bolstering brand association in consumers' memories, consequently increasing product sales through enhanced brand equity, customer loyalty, and relationship marketing.
Menon, 2019.	N/A	Survey	Skincare soap products	Consumers of skincare products	Keller and Aaker's conceptualisation model		Brand awareness and brand image	Brand knowledge, brand recall, and brand association.	Brand salience (Purchase intention)	N/A		CBBE includes brand awareness, physical quality, staff behaviour, and brand image. Customer satisfaction and brand trust also influence building customer loyalty.
Sürücü et al., 2019.	Turkey	Survey	Hotel and restaurant industry	Fast food consumers	CBBE conceptualised a model		Brand awareness, brand image, perceived physical quality	Customer satisfaction, brand trust	N/A	N/A		

3.5 Measuring Brand Equity

3.5.1 Brand awareness

Among the identified measures, most studies have frequently used brand awareness as a measurement or component of brand equity in CBBE measurement development. Initially regarded as a distinct component of CBBE, and particularly a launchpad for building up to other components, Aaker (1991) has idealised brand awareness as the ability of potential consumers or buyers to recognise or recall a brand into specific categories given various brand sets. On this same note, familiarity implies the customer's knowledge and experience with a particular brand and is acknowledged to impact customer behavioural variables significantly. Given the conceptualisations, brand awareness is thus operationalised as an ability to distinguish and recall one brand among many, thus a measure of equity value. Twelve included studies discussed brand awareness, revealing its significant mediator role as a measure of brand equity (Chen & Tseng, 2010; Chen et al., 2019; Han et al., 2015; Hurriyati & Setiawan, 2017; Jeon & Yoo, 2021; Menon, 2019; Muhammad & Muath, 2018; Naggar & Bendary, 2017; Poonam & Dr. Sushil Sharma, 2023; Qiao et al., 2022; Shariq, 2019; Sürücü et al., 2019). For instance, Shariq (2019) found that brand awareness significantly influences brand loyalty, substantially impacting brand equity in the fast-moving consumer goods industry. Shabbir similarly implied that brand awareness fully mediates the influence of brand loyalty and brand image on overall consumer brand equity (Shabbir et al., 2017). Accordingly, Chen & Tseng (2010), in the context of the airline service industry, connote that service quality boosted the influence of consumers' repurchase intention by fostering a positive link between brand awareness and consumer perceived value on brand equity. In the identified measures, brand awareness has been widely used in most studies as either a measurement or as part of brand equity CBBE measurement development. Nonetheless, most of the studies have not provided psychometric validation or cited published scales to measure brand awareness, as evident in the study conducted by Yoo & Donthu (2001) and Netemeyer et al. (2004), demonstrating a deficiency of the available literature concerning standard measure accreditation.

3.5.2 Brand Association/Image

Aaker (1996) defines brand identity as a "unique set of brand associations that brand strategists aspire to create and maintain. Thus, these associations represent what the brand stands for, including a promise to customers". According to Aaker (1991), brand association refers to a customer's memory of a brand. For Keller (1993), brand associations are "the other informational nodes linked to the brand node in memory and contain the meaning of the brand", hence the "image". This implies brand association or image as anything according to the meaning of a brand. In this study, eight articles conceptualised brand associations or image as CBBE's distinct measure, including other distinct brand personality constructs, especially in the hotel and tourism industry (Dada, 2021; Han et al., 2015; H. M. Jeon & Yoo, 2021; Menon, 2019; Muhammad & Muath, 2018; Naggar & Bendary, 2017; Shariq, 2019; Sürücü et al., 2019). Mainly, Han et al. (2015), for example, attribute brand reputation as a crucial subset of brand association, critical in establishing brand trust for overall brand equity. Jeon & Yoo (2021) also attribute brand experience to perceived value as a crucial construct of brand association or image, perceived quality, and brand awareness, which constitute fundamental measures for brand loyalty and attainable overall brand equity in the context of food service grocery consumers. According to Naggar & Bendary (2017), brand loyalty was a product of CBBE measures such as brand association, brand experience and perceived brand quality. In another study, Qiao et al. (2022) acknowledged that even though product perceived value had no formidable direct association with consumer-based brand equity, it positively influences consumer affective commitment and brand resonance, which mediates brand-perceived value association and, ultimately, brand equity. There has been some benefit in that brand association is still one of the leading constructs, although few studies administered validated scales or cite existing frameworks to measure it, which limits the possibility of comparing findings.

3.5.3 Perceived Quality

Perceived quality generally relates to the opinions of consumers and customers on the extent to which a product or service meets their expectations. Parasuraman et al. (1988) attribute perceived quality as a five-dimensional concept of reliability, tangibles, responsiveness, assurance and empathy (SERVQUAL); unlike other scholars who operate the CBBE framework, it refers to operationalising it as a consumer value dimension. However, based on the CBBE framework, it refers to the customer's judgement on a product's overall excellence or superiority. In this regard, it is arguable that perceived quality has no direct relationship with the actual brand or product performance. Among the studies included in this study, ten discussed perceived quality as a dimensional measure of brand equity or antecedent of other dimensional components based on the industry contexts (Chen & Tseng, 2010; Choi et al., 2017; Han et al., 2015; Jeon & Yoo, 2021; Muhammad & Muath, 2018; Naggar & Bendary, 2017; Netemeyer et al., 2004; Poonam & Dr. Sushil Sharma, 2023; Shariq, 2019; Sürücü et al., 2019). Consumers' high perceived

brand equity is believed to increase brand preference, thus bolstering overall brand equity. According to Vera and Trujillo (2017), self-identification with a brand, termed self-congruence, and perceived brand quality are key variable measures for brand loyalty. In the context of the brewing industry, Juga et al. (2018) found that value-adding services were crucial in establishing a company's brand equity via perceived service quality, contributing to its significant dimensional measure. Again, Jeon & Yoo (2021) demonstrated the influence of brand experience on perceived value as an antecedent of the perceived quality measure of consumer brand equity, incorporating the affective, sensory, behavioural and intellectual factors. From a financial perspective, Netemeyer et al. (2004) suggest that perceived quality, alongside perceived value for the cost and uniqueness of a brand, are typically essential direct measures of willingness to pay a price premium for a product, which consequently translates to customers' brand purchase behaviour. Naggar and Bendary (2017) also argued that besides brand associations, the perceived brand quality potentially bolsters brand loyalty by establishing a trustful and reliable association among consumers of mobile services, hence, according to brand equity. Although being widespread, the operationalisation of perceived quality was varied as well, and most of the research did not refer to well-known scales such as SERVQUAL or other scales used basis.

3.5.4 Brand Loyalty

Brand loyalty is defined as "an attained state of enduring preference to the point of determined defence", where the endurance is attributed to consumers purchasing the brand repeatedly, and the point of defence implies the consumers defending themselves from competitor brands. However, brand loyalty has traversed various operational and conceptual crises similar to the above dimensions. Definitions from past scholars reflect overt behaviours in conceptualising brand loyalty, also termed as customer or consumer loyalty, and include intentions and positive word of mouth, repeat purchases, and covert attitudinal behaviours (price sensitivity, liking and preference). The initial operationalisation of the CBBE framework failed to capture the attitudinal and behavioural nature of loyalty, defining brand loyalty as "the attachment that a customer has to a brand" (Aaker, 1991). Recent studies have, however, included satisfaction and price premium as indicators or measures of brand loyalty. In this review, 17 studies operationalized and conceptualized brand loyalty and its antecedents as crucial measures of brand equity (Bose et al., 2020; Chen & Tseng, 2010; Choi et al., 2017; Dada, 2021; Ebrahim, 2020; Hurriyati & Setiawan, 2017; Jeon & Yoo, 2021; Juga et al., 2018; Kressmann et al., 2006; Mabkhot et al., 2017; Muhammad & Muath, 2018; Naggar & Bendary, 2017; Ong et al., 2015; Poonam & Dr. Sushil Sharma, 2023; Şahin et al., 2011; Shariq, 2019; Vera & Trujillo, 2017). Findings by Choi et al. (2017) extrapolate that brand personality and brand experience directly influence brand prestige, which consequently impacts the quality of brand relationships and behavioural attitudinal loyalty. In the context of the hotel and tourism industry, Sürücü et al. (2019) claim that enhanced customer satisfaction and brand trust are significant measures worth considering while building customer loyalty as an overall measure or dimension for CBBE. On the same course, Sahin et al. (2011) allude that brand trust, experience and satisfaction certainly positively influence brand loyalty as a measure of brand equity. In the assessment of factors that influenced decisions for visitation to a particular tourist site, Hurriyati and Setiawan (2017) established that destination personality, despite positively correlating with brand attractiveness, is both a critical measure of brand loyalty since it positively influences customer loyalty to the brand, and thus overall brand equity. Research by Ong et al. (2015) also ascertained that brand experience is probably a measure of brand equity through its critical mediating influence on behavioural and attitudinal brand loyalty. Even though brand loyalty was the most frequently addressed indicator, in less than half of the papers, there was the use of validated scales or declarations to original works like that of Aaker (1991) or Keller (1993) in operational definitions.

3.5.5 Structural Associations among Brand Equity Components

Although the CBBE conceptualisation by Aaker (1996) does not suggest any imminent relationship between the dimensional constructs, awareness was implied as the primary measure for the other components since it influenced attitudes and perceptions. Keller's (1993) model alternatively presented a pyramid with brand salience at the base, followed by brand performance and imagery, brand judgments defined by brand performance, consumer effect defined by imagery, and consumer brand resonance. However, there needed to be an operationalised clear relationship among the core equity components. In this regard, researchers have revealed various structural correlations among these CBBE model dimensions. For instance, loyalty is considered an outcome or a component of CBBE. Concerning Shabbir et al. (2017), findings reported that brand awareness fully mediated the influence of brand loyalty on brand equity, implying that brand loyalty was an outcome of brand awareness. In the article by Bose et al. (2020), brand equity is perceived as a CBBE component mediated by place brand salience, engagement, and perceived quality to boost brand equity. In the investigation by Mabkhot et al. (2017), a quality relationship was found between brand loyalty and brand image. Brand trust was also noted as a measure of brand personality and brand loyalty, depicting the latter as distinct customer brand equity components.

Ong et al. (2015) also conceptualised brand loyalty as a CBBE component by reporting that brand experience mediated behavioural and attitudinal loyalty facets of brand equity. The same was the case in other studies, which acknowledged the structural relationship of CBBE measures by considering brand loyalty as an integral component, alluding to the mediating roles played by brand trust, experience, destination personality, brand attractiveness and satisfaction with brand loyalty (Hurriyati & Setiawan, 2017; Sahin et al., 2011).

4. Discussion

Our analysis reveals ongoing challenges in brand equity research, particularly regarding standardised definitions and measurement approaches across different contexts and industries (Buil et al., 2008; 2013; Veloutsou et al., 2013; 2020). Despite brand equity being a key issue in marketing, its conceptualisation still needs to be revised due to the contradictory viewpoints on its theoretical framework and dimensional building blocks that act as its antecedents and measures as an intangible marketing asset (Ishaq & Maria, 2020). Whereas consumer-based brand equity (CBBE) is of most interest in terms of brand equity to most studies because of its implications on strategies of company behaviour, financial-based brand equity (FBBE) is of significant consideration in determining the conversion of brand value into firm performance and shareholder value. A contrasting take on the integration of CBBE and FBBE would render a greater picture of brand equity, particularly when it comes to resonating the resonance of marketing with financial results. Current research predominantly focuses on consumer-based measures, with limited integration of financial perspectives in brand equity assessment. As an example, the Interbrand brand valuation method, or a brand contribution ranking view, is financial-based in nature and offers a quantitative, balance sheet-oriented analysis. CBBE, on the contrary, is more dependent on the perceptions of the consumer. A healthy debate on brand equity should focus on both sides to make it more theoretical and practical.

Brand equity presents both perceptual and financial dimensions. However, perceptual equity has stood as a proxy for financial equity since most studies have focused on establishing the influence of most perceptual indicators in different industrial brand contexts. This one-dimensional CBBE approach to dimensions - brand awareness, loyalty, perceived quality, brand associations - confines us to the understanding of the contribution of intangible perceptual factors to tangible financial performance. Also, it can be emphasised that the discussion of brand equity can be enhanced by incorporating more general theoretical developments that started to guide the resources-based view (RBV) and signalling theory. Within the RBV, brand equity can be viewed as a strategic intangible resource which is in the form of a sustainable competitive advantage. Through a signalling theory perspective, dimensions of brand equity are plausible indicators of product quality and reliability of the firm, which affects consumer choice in the presence of asymmetric information. Such theories fill the gap between the perceptual and the financial strategies towards brand equity.

This study, however, reveals that the multifaceted brand equity elements and their interrelations are built and measured in numerous diverse ways based on the context of the brand product or service industry. As such, there should be standardisation to develop a robust measure of brand equity for each product or brand context. To overcome these shortcomings, this paper postulates a consolidated brand equity model which integrates both CBBE and FBBE based on a dual lens; the consumer perception and the financial perspective lens, which is based on RBV and signalling theories. Having such a framework can help coordinate the evaluations of brands by marketing people and financial managers in an evidence-based approach. Towards this call, this suggests deploying uniform CBBE dimensions, measures and structural relationships. However, brand awareness is considered brand familiarity, brand association or brand image, brand loyalty, perceived quality or consumer value are considered the complete CBBE components, and their relations and antecedents can be standardised as a measure for brand equity.

The first limitation of this study is that it was only possible to include some of the identified resources on this topic, limiting the robustness of this analysis. Again, despite following the PRISMA guidelines, the subjective bias during the process might have led to a lack of critical literature that could be insightful for this topic. There needed to be more evaluation of evidence quality, and this study only provided a descriptive summary of the measurements identified from the included articles, implying a lack of reference accreditation and validation standards for each identified measure or construct of brand equity.

5. Conclusions and Future Directions

Brand equity is considered an essential asset for any industry; hence, this critical review aims to highlight this concept and mainly review variables of its conceptualization, dimensions and measurements and, finally, the structural interrelationships presented by its components, the recent business competition, firms have recognized the importance of understanding or establishing brand equity to develop and enhance competitive advantage in the business competitive world. This review summarises empirical studies attributing brand equity to various

consumer value propositions or marketing strategies. This study provides a comprehensive overview of brand equity components and their interrelationships, offering practical insights for brand management strategies.

This study advocates for future research that needs to distinguish the crucial measures of financial-based brand equity from consumer or customer-based brand equity. This is so because, despite FBBE and CBBE being related or consummate concepts, CBBE tends to focus on consumer or customer perspectives, measured by perceptual elements. At the same time, FBBE attributes to the financial basis that depicts a brand's monetary value from the equity components. Moreover, this study identifies a close relationship between brand equity, its components, antecedents, and consequences. Strong brands typically demonstrate positive reinforcement cycles where high equity drives further brand strength, while weak brands may experience declining equity over time. The latter presents a significant research gap that needs to be addressed by future studies evaluating such feedback's impact on product value or brand equity.

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The data that support the findings of this study are available on request from the corresponding author. The data are not publicly available due to privacy or ethical restrictions.

Data sharing statement

No additional data are available.

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